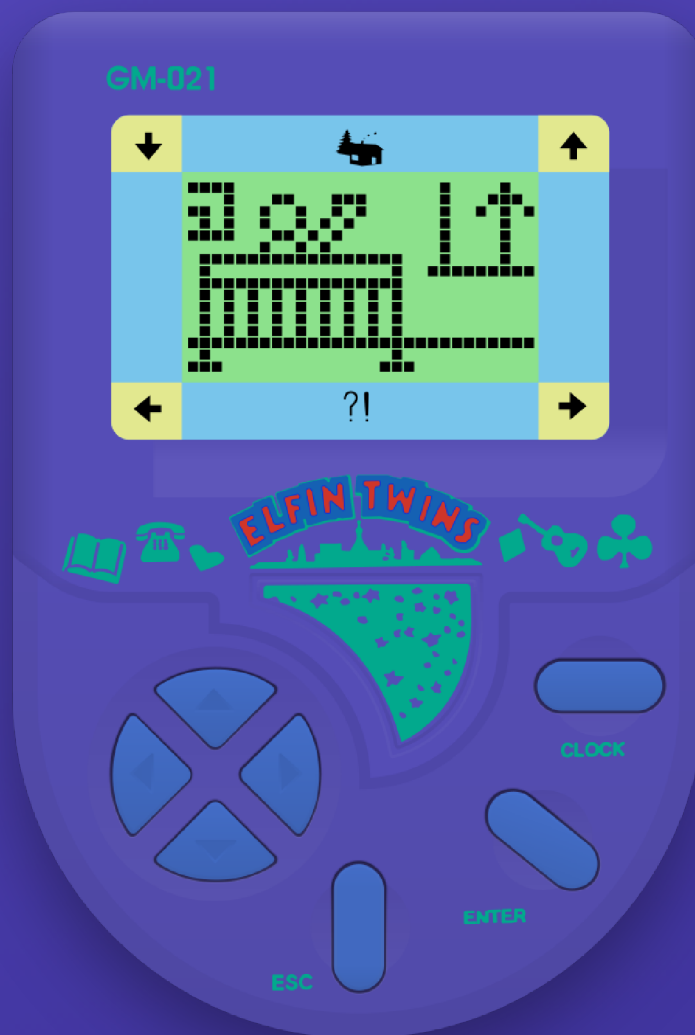


GM-021

Elfin Twins

Instruction manual for the virtual pet
and its Game Boy Advance edition



An unofficial manual. No original manual was available, so everything in it was reconstructed from the toy's own program: it was run on an emulator, its screens were recorded and its code was read. All screen pictures are taken from the real program.

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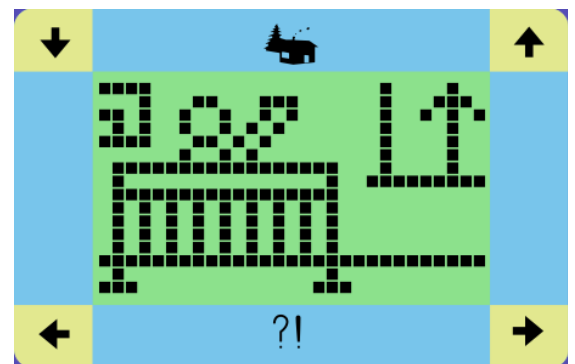
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1. Welcome to Elfin Twins

Elfin Twins is a pocket virtual pet from the late 1990s. Instead of one creature you look after a pair of little elves, the **twins**. They arrive at the end of a short story, grow up from babies to adults over about three and a half weeks, live in a house at the edge of a town and like to go out: to the town and, when they are older, to the running track.

You feed them and give them drinks, keep them clean, put them to bed, call the doctor when they are ill and keep them busy with games, outings and TV. Everything you do changes their mood, intelligence, stamina, friendships, weight and money, and you can check all of these on the status meter.

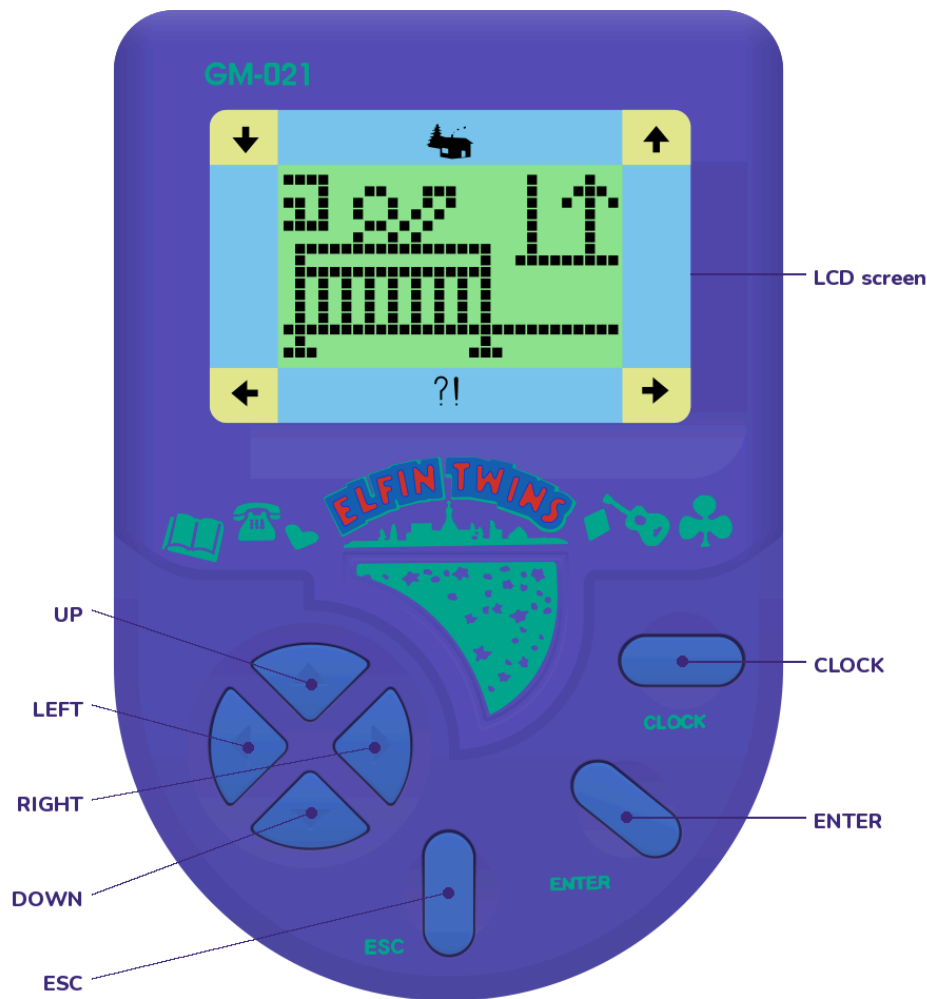
With a second toy you can connect the two and let the twins play tug-of-war, rock-paper-scissors or arm wrestling against a friend's twins.



The twins at home (babies, first day).

About this manual. Each rule is marked with the source it was taken from. Most come straight from the toy's program, others from watching the emulated toy. Where the purpose of something could only be guessed, the manual says so.

2. Getting to know your toy

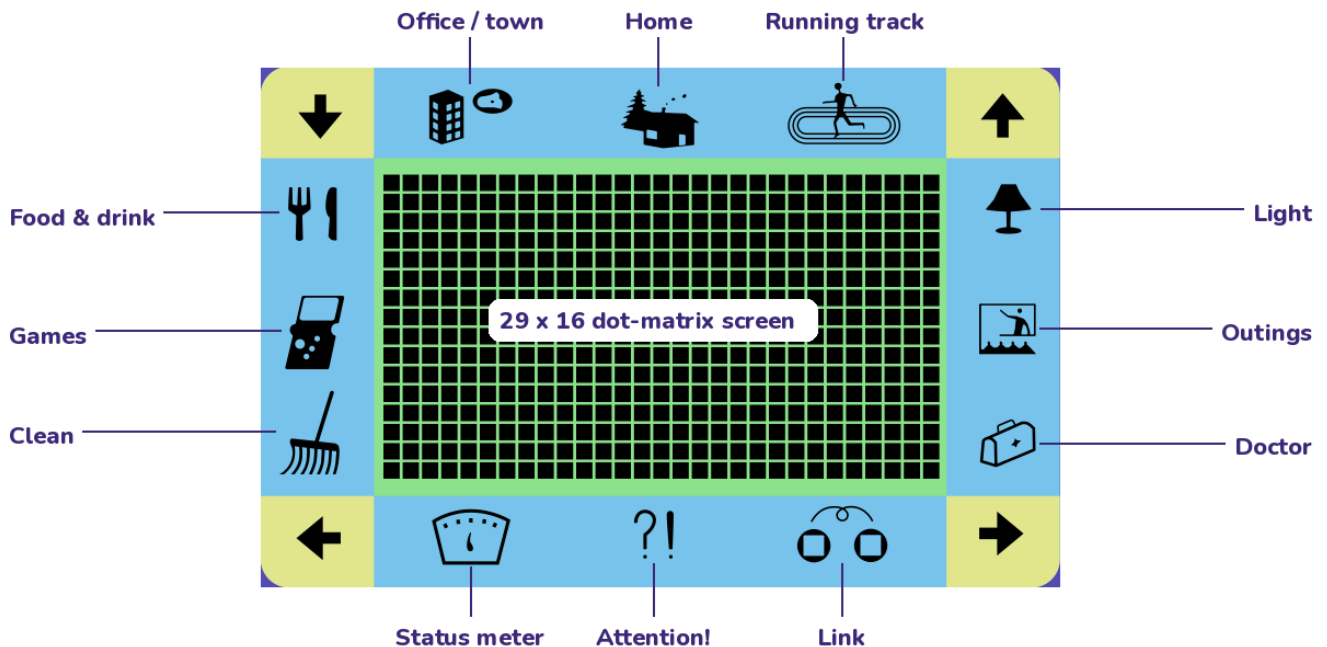


Button	What it does
◀ ▶	Move the menu cursor, browse choices, change values.
▲	Choose where the twins go (home, town, running track).
▼	Cancel the choice of destination.
ENTER	Select / confirm. Held down for about 2 s it is a "long ENTER".
ESC	Go back / leave a screen.
CLOCK	Show the clock. Hold it for about 2 s to stop and set the clock.
ESC + ENTER	Pressed together: emergency rescue, or start a new pair of twins (see chapter 17).

Inside the toy. The toy has only five button contacts. ▲ presses LEFT and RIGHT together and ▼ presses CLOCK and ENTER together, and the program tells them apart by which contacts close. A tiny buzzer plays the tunes and a two-wire socket connects two toys.

3. The screen and its icons

The middle of the screen is a grid of 29×16 dots where the twins, their home and all the animations appear. Around it are 16 printed icons that light up when needed. The picture below shows every icon lit at once, which never happens in normal play.

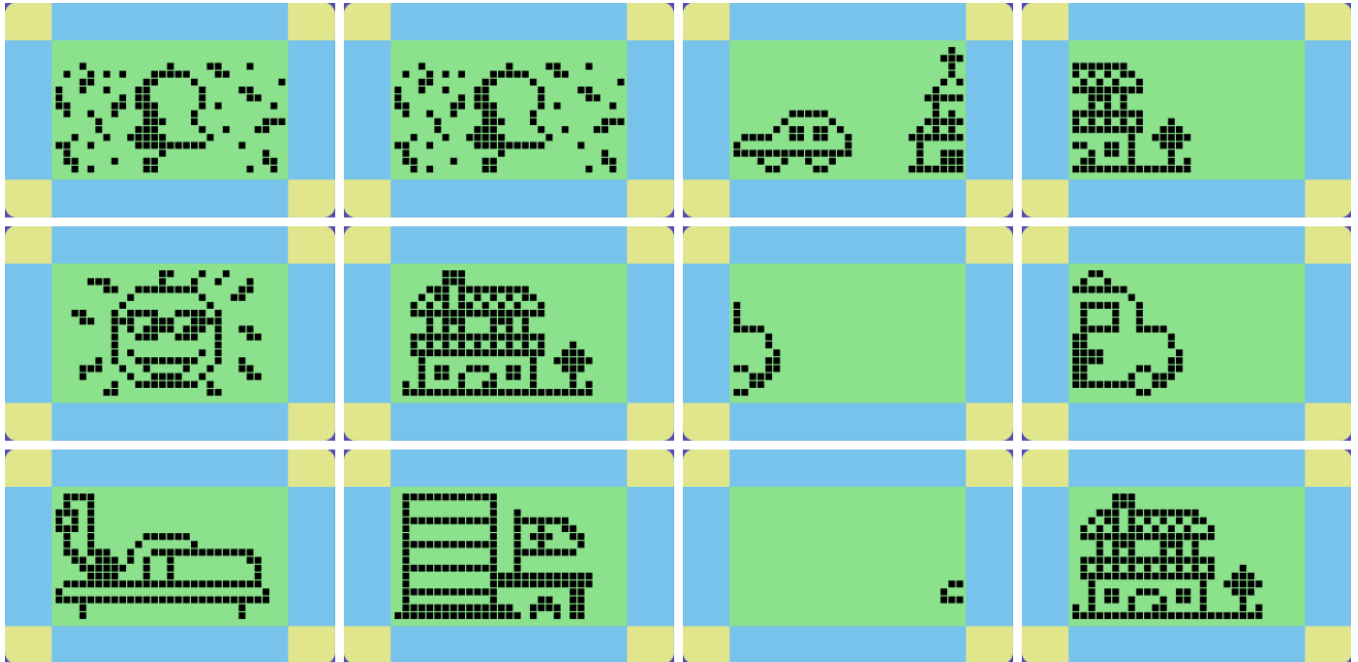


Where	Icons	Meaning
Top row	Office / town · Home · Running track	Where the twins are, and their destination while you choose one.
Left column	Food & drink · Games · Clean	The eight menu icons . One is lit while you move the cursor.
Right column	Light · Outings · Doctor	
Bottom row	Status meter · Link	
Bottom middle	?!	The twins need something (chapter 14).
Corners	▼ ▲ ◀ ▶	Which buttons can be used right now.

4. Getting started

The story

When batteries go in (or after a reset) the toy tells a short picture story that lasts about two and a half minutes. It ends with the twins arriving in their new home as babies. The buttons are not used during the story, so just watch.

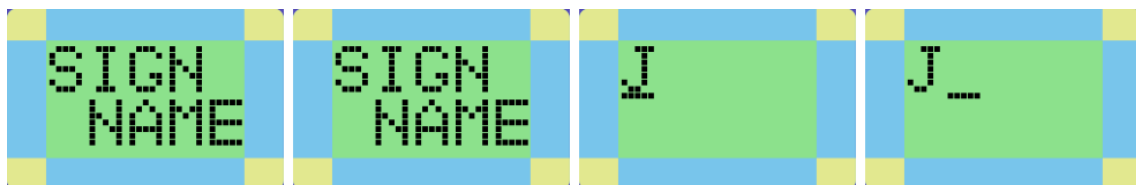


Scenes from the opening story.

Signing your name

At the end of the story the toy asks you to **SIGN NAME**: you can sign with up to 10 letters.

- /  choose a letter (A–Z, a space and a few symbols).
-  accepts the letter and moves on to the next one.
-  finishes. If you do nothing, the twins move in after about half a minute anyway.



SIGN NAME and the first letter.

Setting the clock

The toy starts at **12:00**. Set the real time, because the twins' day (mealtimes, bedtime, going out) follows the clock.

- 1 Press **CLOCK** to show the clock.
- 2 Hold **CLOCK** for about 2 seconds. The screen shows **CLOCK STOP** and time stands still.
- 3 Press **ESC** to start setting the time.
- 4 ◀ adds one hour, ▶ adds one minute. Keep pressing until the time is right.
- 5 Press **ENTER**. The clock starts at 0 seconds and the password screen **NEW P.W.** appears (chapter 19). Leave it at **00** for no password and press **ENTER**.

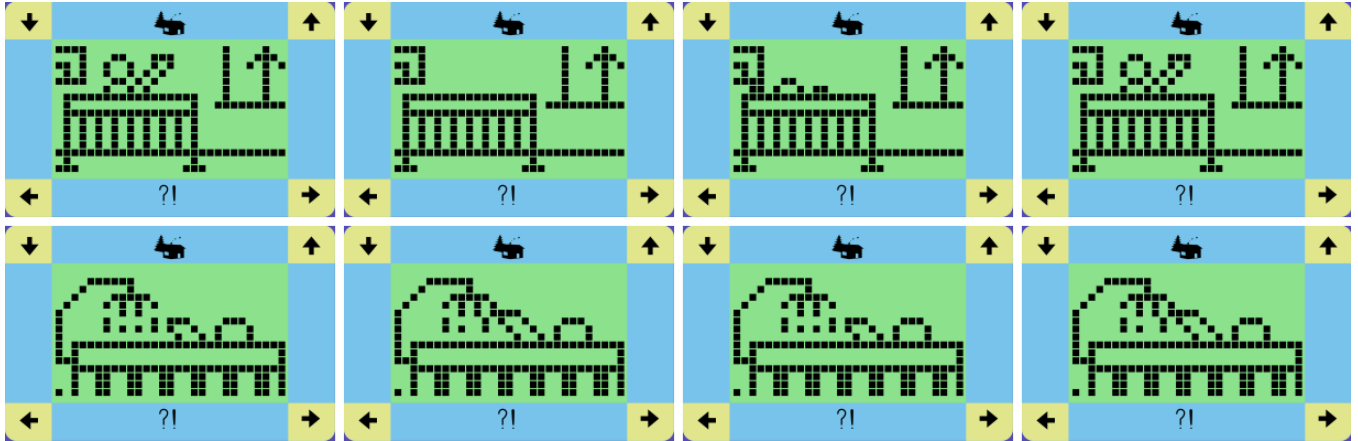


Clock screen (with the sound bell), sound off, CLOCK STOP, NEW P.W., password digits.

Tip: "CLOCK STOP" also works as a **pause**. While it is shown, time does not run and the twins do not get older or hungrier.

5. The home screen and the menu

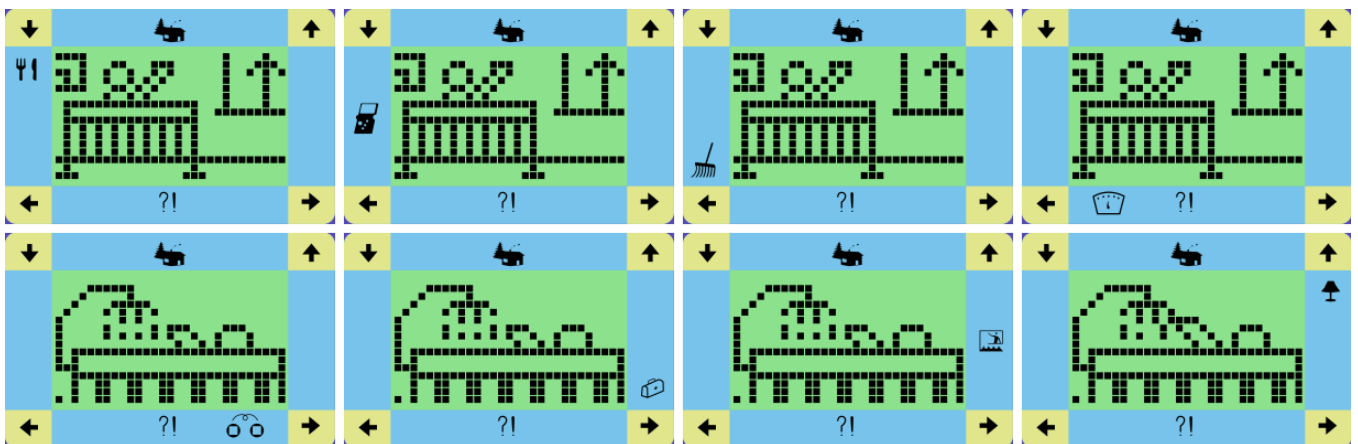
Most of the time you see the twins in their surroundings, going about their day. What they do depends on their age, where they are and the time of day. Now and then they call for attention with **?!** and act out what they need: yawning when sleepy, a mess when things need cleaning, or looking poorly when they are sick.



A few moments at home.

Using the menu

Press **◀** or **▶** and one of the eight menu icons lights up. Each press moves to the next icon, round the screen in this order:



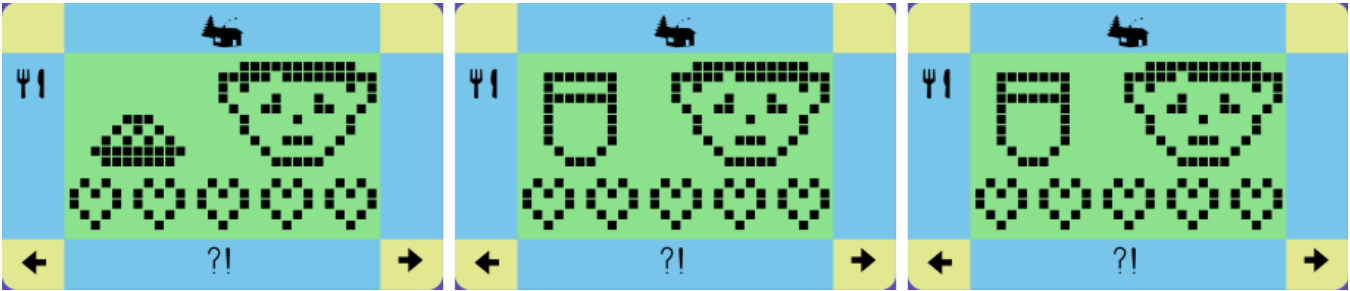
Food & drink → Games → Clean → Status meter → Link → Doctor → Outings → Light.

Press **ENTER** to open the lit icon. Inside a screen, **◀**/**▶** browse, **ENTER** chooses and **ESC** goes back. If you leave the toy alone for two minutes inside a screen, it goes back to the home screen by itself.

Tip: While the twins are in the middle of an animation, a button press is sometimes ignored. Just press it again.

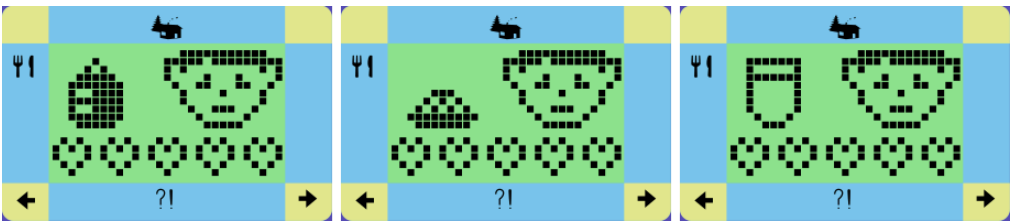
6. Food and drink

Open **Food & drink** (knife and fork). Three things to eat or drink are offered and the screen moves from one to the next every few seconds. Wait for the one you want, or pick it with ◀/▶, then press **ENTER**. Press **ESC** to leave.



The three choices for babies at home.

Choice	What it does
Left and right: meals	Fill the tummy and give + stamina . Some meals also add weight. When the tummy is full (after about five meals) the twins refuse more food.
Middle: drinks	Quench thirst and give + stamina . Always accepted. For older twins at home the drink is a stronger one (probably coffee): + IQ - mood , and it costs \$100.



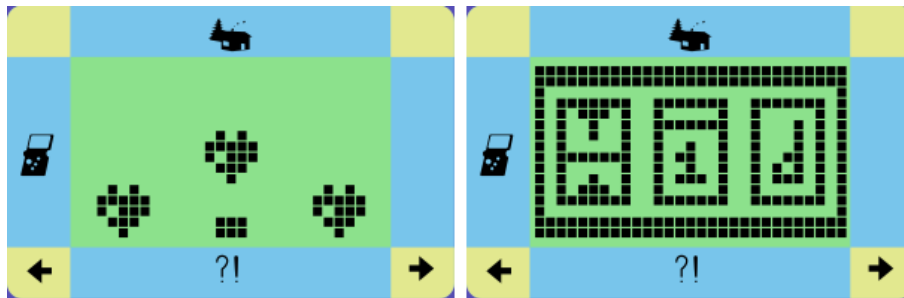
Dinner time.

What is offered depends on the twins' age (babies and children, or teenagers and adults) and on whether they are at home or out in town. At the running track they refuse to eat.

Tip: In the morning the twins wake up with an empty tummy and thirsty, so give them breakfast and a drink. Over the day they digest, and a digested meal eventually means a mess to clean.

7. Games

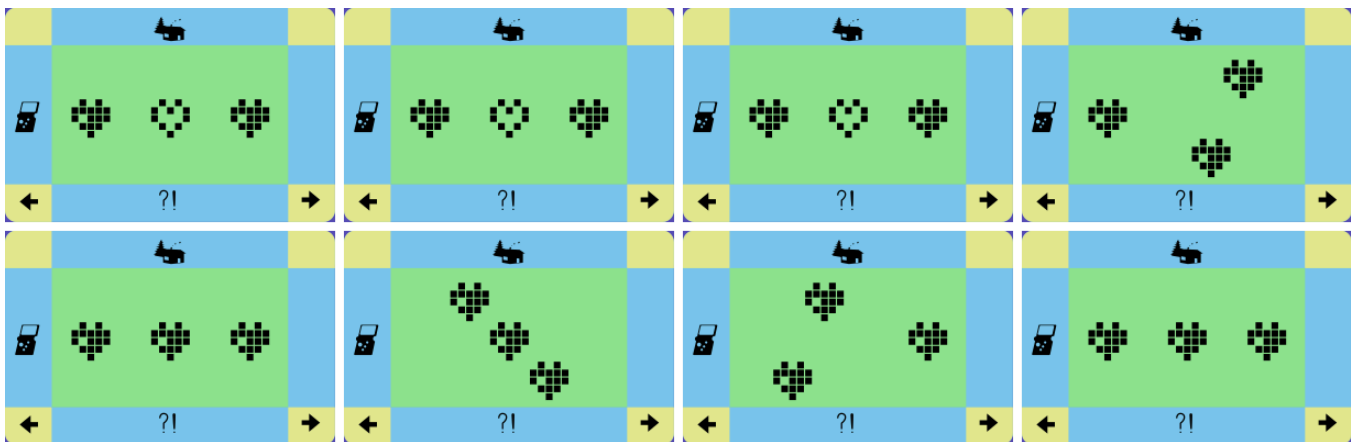
Open **Games** (the little handheld console). Choose one of the two games with ◀/▶ and press **ENTER**.



The two games: find the heart, and the lucky reels.

Find the heart

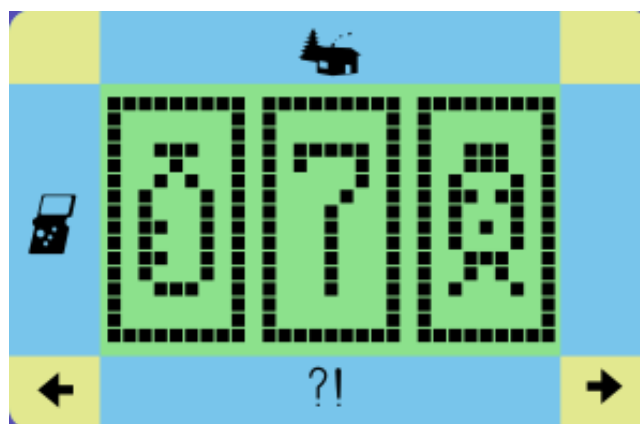
One of three hearts is hollow. Watch it, because the hearts are shuffled. When the arrow appears, move it under the hollow heart with ◀/▶ and press **ENTER**.



The hearts are shuffled; keep your eye on the hollow one.

Lucky reels

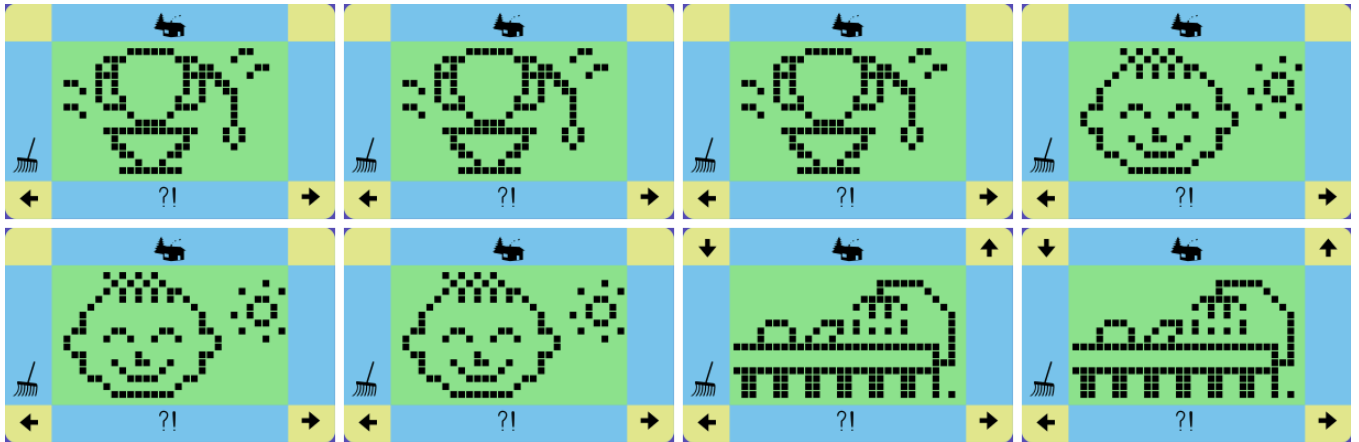
Three reels spin like a slot machine. Press **ENTER** to stop them.



Games mostly affect **IQ** and **stamina**. In our tests a round raised or lowered them depending on the result.

8. Cleaning

Open **Clean** (the rake) when the twins or their room are dirty: after they have digested a few meals, or when the room is messy. They get a bath and are happy again. If nothing needs cleaning they just say no.

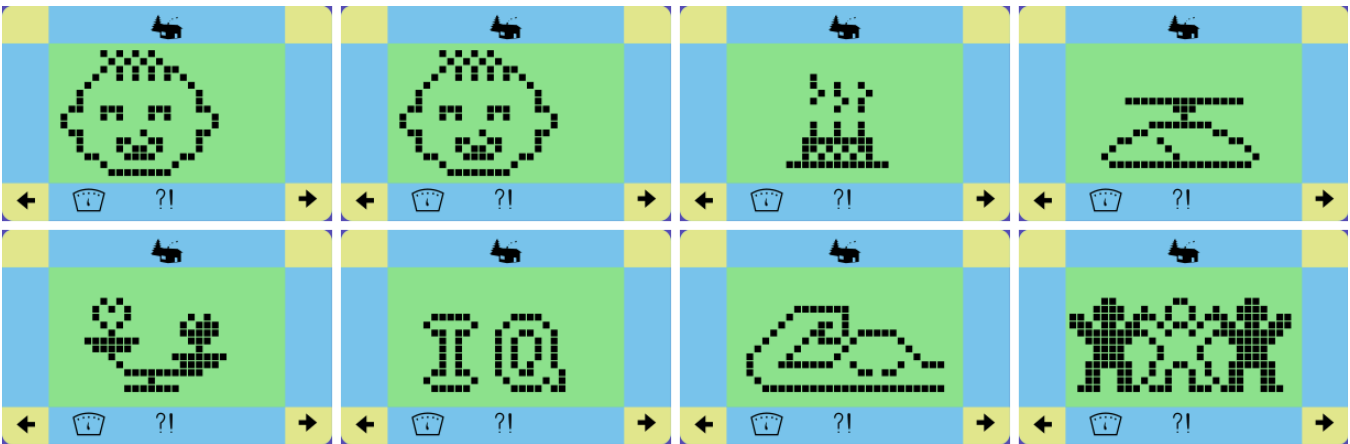


Bath time.

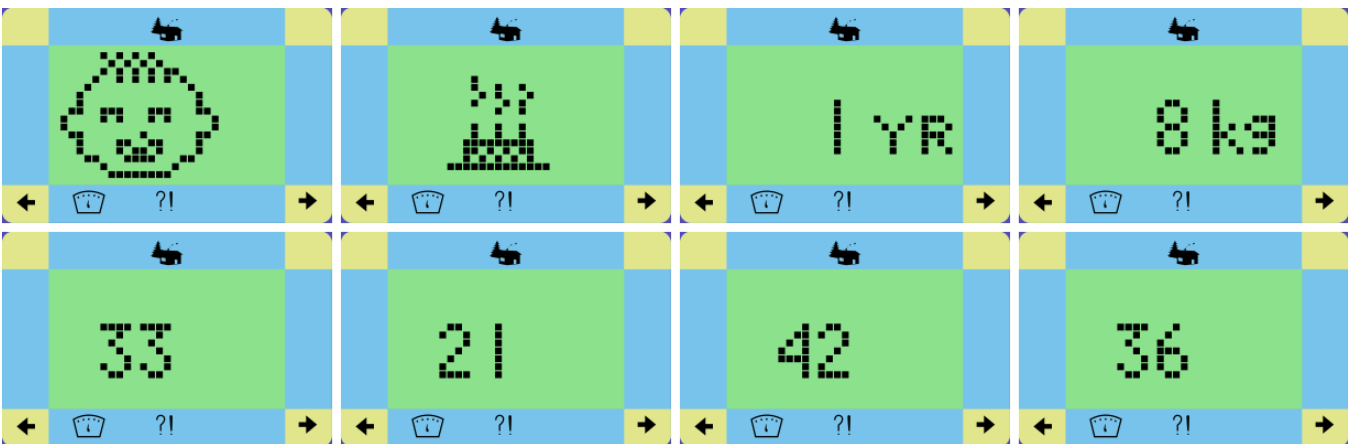
Leaving a mess costs money (\$200 each time it builds up) and makes the twins ill.

9. The status meter

Open the **Status meter** (the dial). Its eight pages come up one after another. Browse with ◀/▶, press **ENTER** to see the value, and **ESC** to leave.



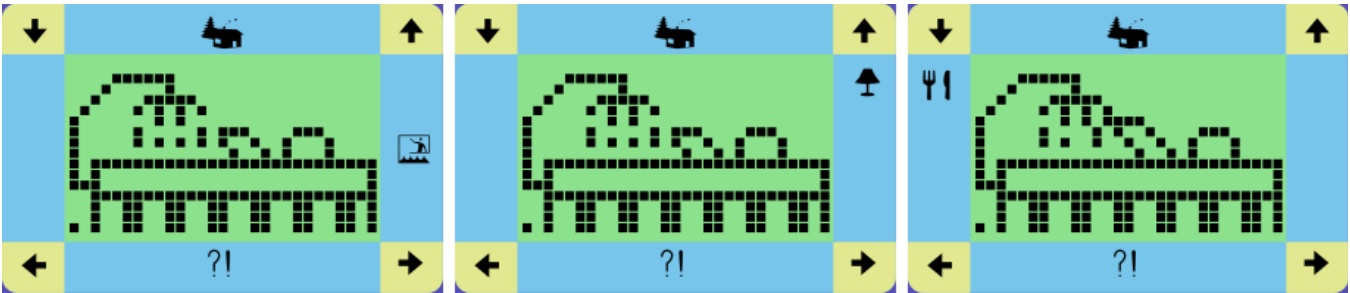
Page	Shows	Notes
Baby face	Growth	Adults show a special portrait here.
Birthday cake	Age	Shown as "YR" but counted in days : one toy day is one "year".
Scales	Weight in kg	Starts at 8 kg. Meals add weight; hunger and late bedtimes reduce it.
Flowers	Mood	Shown as 0–99. Each drops by one point a day on its own; raise them with games, outings and play.
IQ	Intelligence	
Sneaker	Stamina	
People	Friendship	
Banknote	Money	Starts at \$2400 (chapter 20).



Values for a new pair: age, weight 8 kg, mood 33, IQ 21, stamina 42, friendship 36, money \$2400.

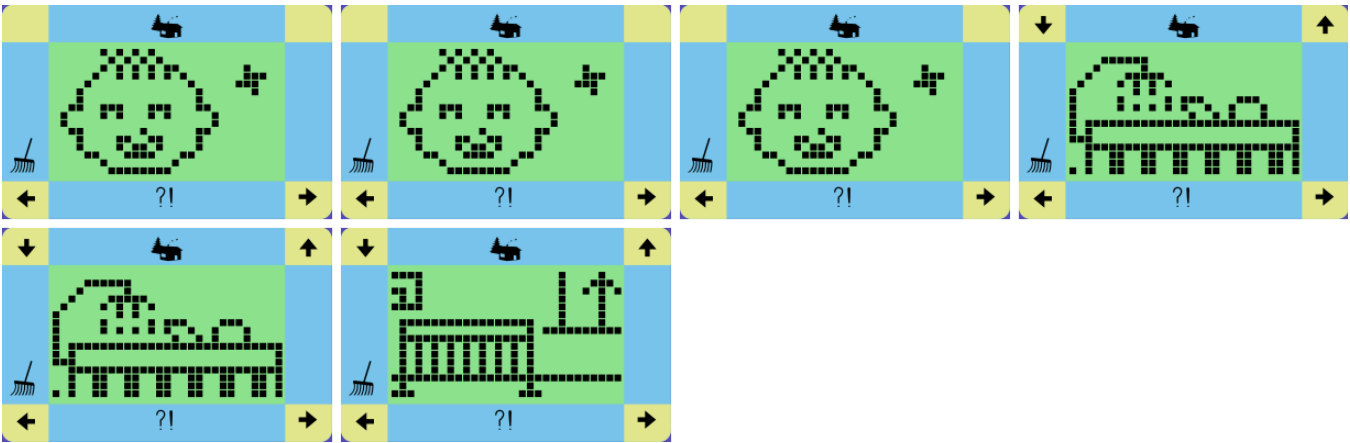
10. Outings

Open **Outings** (the little picture of someone at the seaside) to give the twins something to do. Pick an activity with ◀/▶ and press **ENTER**. At home the choices are:



Watch TV · Listen to music · Read and learn.

Activity	Effect
TV	+ friendship - \$100
Music	+ mood - \$100
Reading / learning	+ IQ - friendship



Music makes the twins happy.

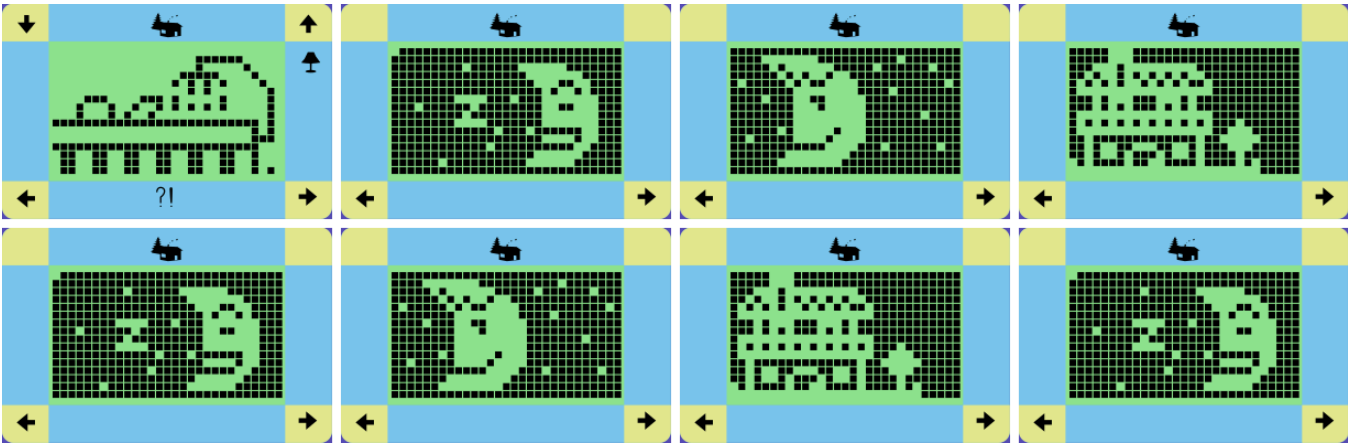
Older twins, and twins out in town, get other activities. The program has twelve in total. We could measure their effects but not all their names, so the names below are our best guesses:

Kind of activity	Effect
Work (three kinds)	+ money and + IQ , + friendship or + mood
Sport (three kinds)	+ stamina , and the tummy or thirst goes down
Hard work	+ \$300 - stamina
Party	+ 2 mood + 2 friendship - \$400
Lessons	+ 2 IQ - \$200

Some activities include a small challenge. Get it right to earn the benefit; get it wrong and it costs \$100.

11. Light and bedtime

From **10 pm** (22:00) the twins get sleepy. They yawn and **?!** lights up. Open **Light** (the lamp) and press **ENTER** to turn off the light and put them to bed. During the day the lamp does nothing.



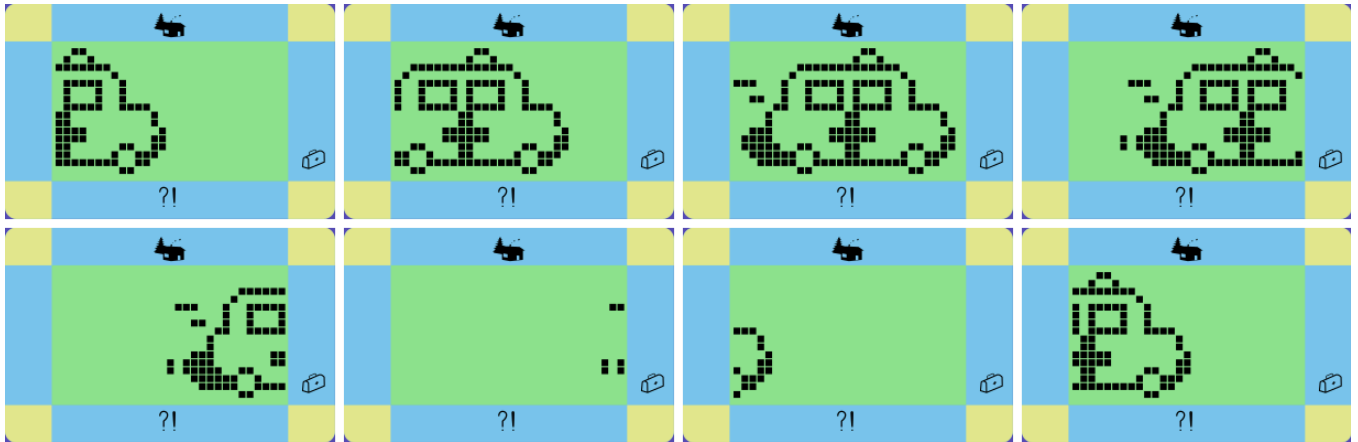
Lights out: the twins sleep under the moon and stars.

Time	What happens
from 22:00	Sleepy: "?!" and yawning. Time to turn off the light.
23:00	If you have not put them to bed they fall asleep by themselves, but lose 1 kg (a late night is not good for them). Putting them to bed yourself gives +1 kg instead.
07:00	The alarm rings and the twins stretch.
08:00	They are up with an empty tummy, thirsty and 1 kg lighter, ready for breakfast.

Tip: The sleeping hours really are sleeping hours: at night the twins do not need anything, so their day starts fresh in the morning.

12. The doctor

Twins who go hungry or live in a mess become ill. They look poorly and **?!** lights up. Open the **Doctor** (the medical bag) and an ambulance takes them to hospital, where they are cured.



The ambulance comes for the twins.

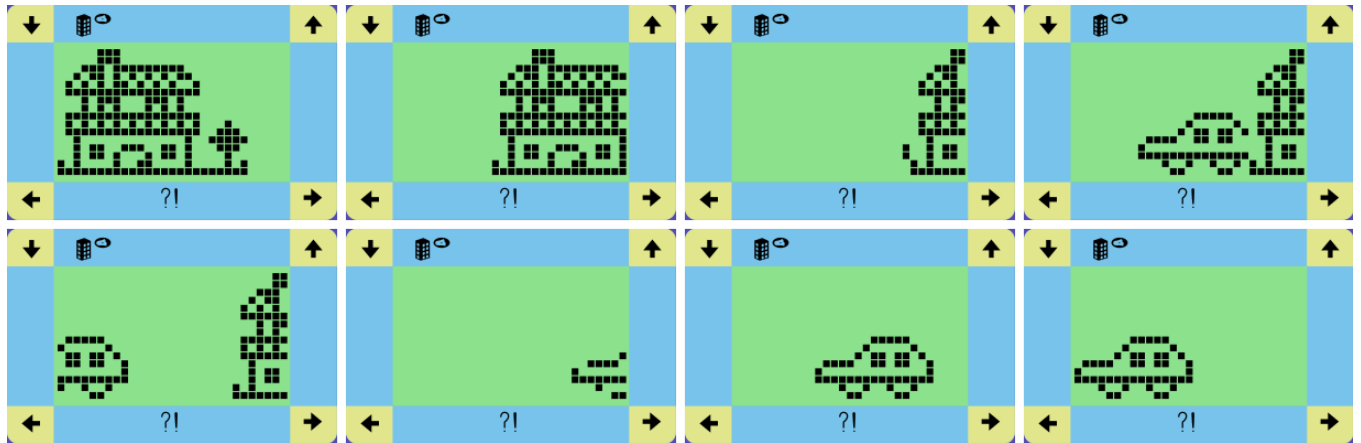
Treatment costs \$100 (\$200 if they were very ill) and the twins come back a little weaker: stamina, IQ and weight drop a bit, and after a serious illness mood does too. If they are not ill, the doctor is not needed and they say no.

Illness gets worse every hour and costs money. If it is left too long, the twins collapse (chapter 17).

13. Going out

The twins can live their day at **home**, go to **town** (the office building icon) and, once they are no longer babies, go to the **running track**.

- 1 Press **▲**. The top row starts to blink.
- 2 Choose the destination with **◀**/**▶**.
- 3 Press **ENTER** to go. **▼** cancels.



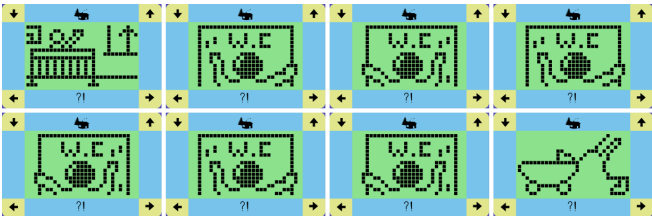
Off to town by car.

Where the twins are changes what they do and what the menus offer (food is different in town, for example, and nobody eats at the running track). After 6 pm (18:00) the only destination is home, and at 8 pm (20:00) twins who are still out come home by themselves.

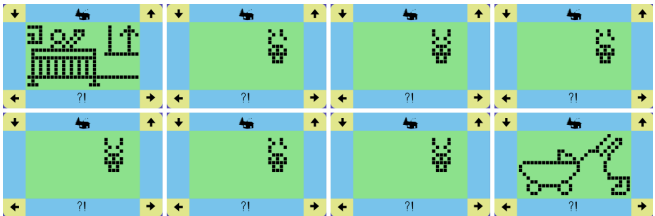
14. The attention call "?!"

The ?! icon lights up, with a beep when it first appears, whenever at least one of these is true:

Reason	What to do
Hungry (empty tummy)	Food: a meal.
Thirsty	Food: the middle choice (a drink).
Dirty, or the room is messy	Clean.
Ill	Doctor.
Money below \$1000	Earn money with outings (work).
Bedtime (after 22:00)	Light off.
Just woken up (7–8 am)	Wait a moment, then breakfast.



Something needs cleaning.



The twins are not well.

15. A day in the life of the twins

The twins' needs change with time. This is how the program ages them:

How often	What changes
Every 30 minutes	They get a little thirstier and lose a little stamina.
Every hour	A meal is digested and a little mess builds up. With an empty tummy they lose stamina and weight and start to get ill instead.
When messy	Each time the mess builds up it costs \$200 and makes them a bit more ill.
While ill	Every hour: a bit more ill, a little weaker in every way, and \$200 goes to medicine.
Every day	A birthday: they are one "year" (day) older, and mood, IQ, stamina and friendship each drop by one point.
Once a day	For a while the room gets untidy and needs cleaning.

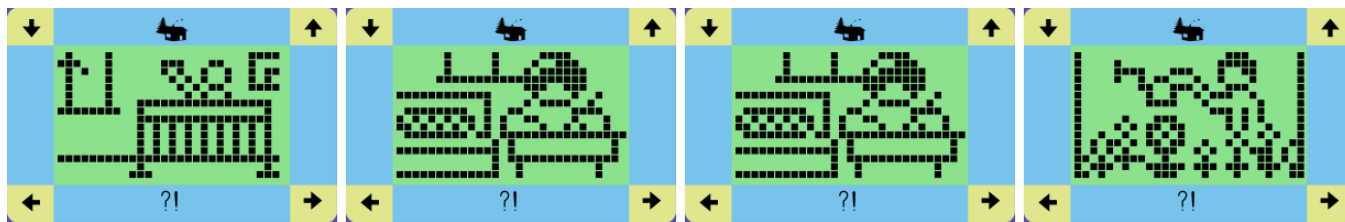
A good daily routine

- 1 8:00** Breakfast: a meal and a drink.
- 2 Morning** Go to town or the running track, or play a game or have an outing.
- 3 Midday and afternoon** A meal and a drink every few hours. Clean when "?! " asks for it.
- 4 Evening** Home before 20:00, then music, TV or games.
- 5 22:00** Light off.

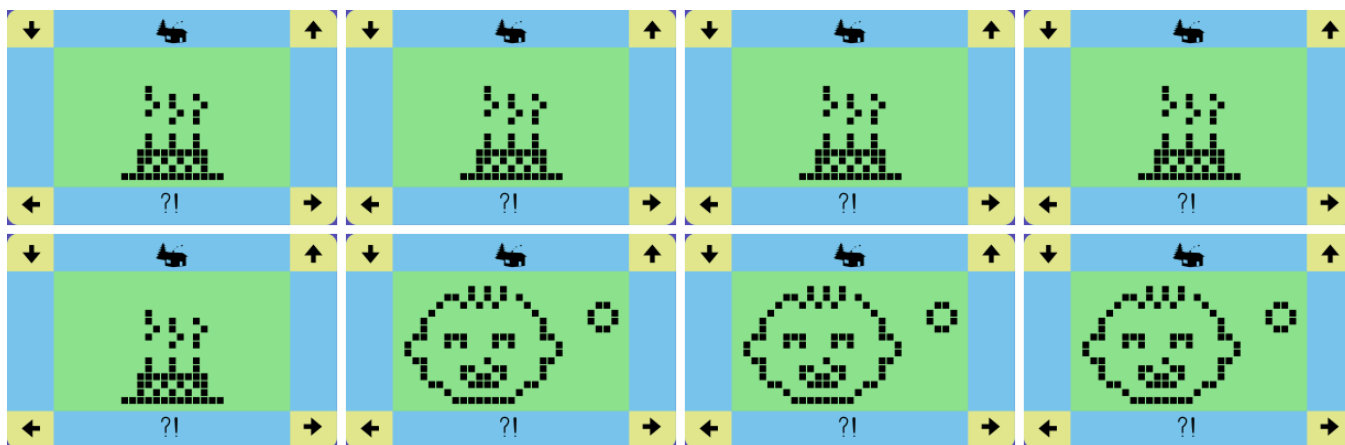
Tip: Check the status meter every day. Mood, IQ, stamina and friendship only go up if you do something for them.

16. Growing up

Every 24 hours, counted from the moment the twins arrived, they have a **birthday**. A cake appears and they grow a day ("year") older. As they get older they move on through four stages, and their home, their games and what they do change with them:



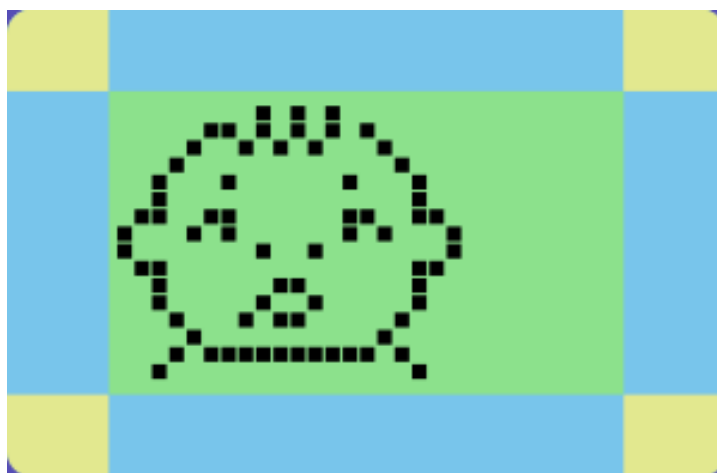
Babies (days 1–3) · Children (days 4–8) · Teenagers (days 9–15) · Adults (day 16 onwards).



Happy birthday!

How long do the twins stay?

When the twins arrive, the toy decides how many days they will stay: **23 to 26 days**. When that day comes the story ends with a farewell portrait of the twins as adults.

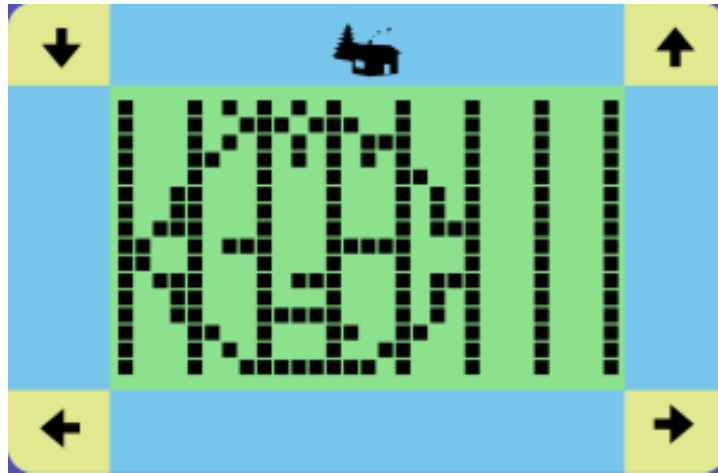


The farewell.

Press **ESC** + **ENTER** together to welcome a new pair of twins. The clock keeps its time.

17. When things go wrong

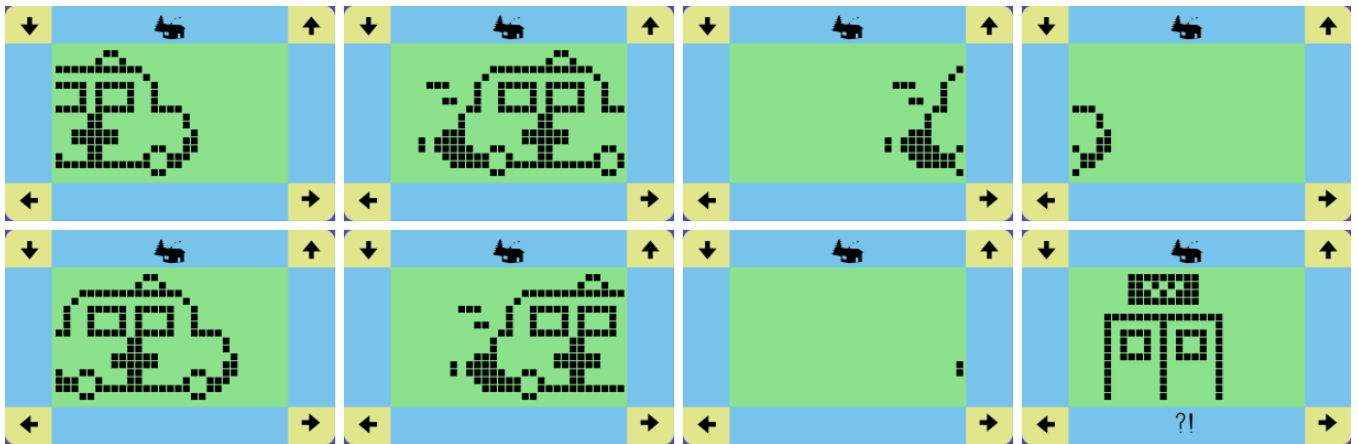
If the twins stay ill for too long (hungry, dirty, never visited by the doctor) or if the money runs out, they **collapse**. The screen shows them lying down and no menus work.



The twins have collapsed.

Emergency rescue

Press **ESC** and **ENTER** at the same time. The ambulance takes them to hospital and they come back healthy. The rescue has a price: their money is reset to \$2000 and mood, IQ, stamina and friendship each drop by 7 points.



Rescued!

24 hours: if nobody rescues them within a day, the twins are gone for good. Then **ESC** + **ENTER** starts a new pair.

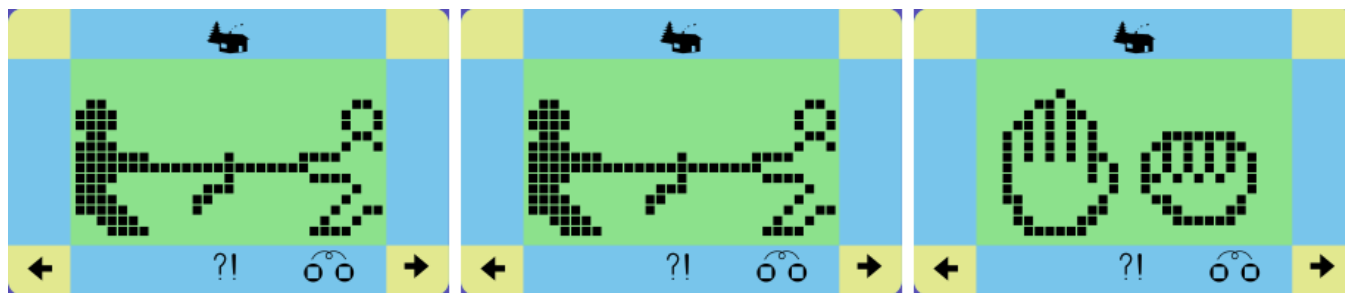
If the buttons seem to do nothing

- The twins may be in the middle of an animation: press again.
- At night they are asleep and most things wait until morning.
- If a password is set, the toy may be asking for it (chapter 19).
- "CLOCK STOP" may be on: finish setting the clock with **ESC** ... **ENTER**.

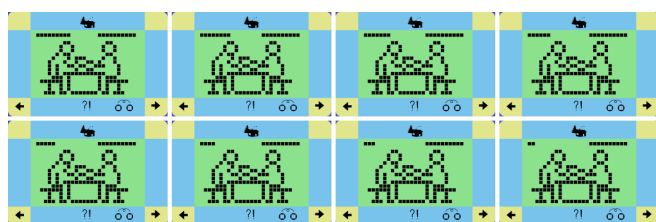
18. Playing with a friend (link)

Two Elfin Twins can be connected with a link cable, so the twins can play against a friend's twins. The toy calls this "linking", and the Link icon shows two figures joined by an arc.

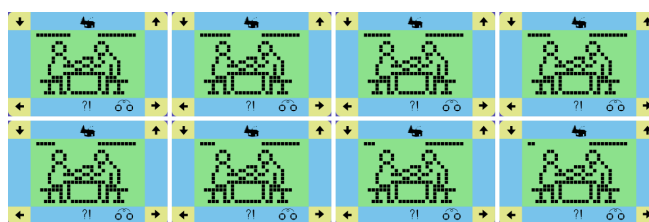
- 1 Connect the two toys (or two Game Boy Advance systems, see chapter 21).
- 2 On **one** toy, open the **Link** icon and press **ENTER**. It calls the other toy.
- 3 The other toy answers **by itself** and both show the link screen.
- 4 On the calling toy choose a game with **◀/▶** and press **ENTER**. Both toys play it together.
- 5 After a game the link screen comes back. Choose another, or press **ESC** to end.



Tug-of-war · Rock-paper-scissors · Arm wrestling.



Tug-of-war on the calling toy...



...and on the friend's toy at the same time.

Playing together raises **+ IQ** **+ stamina** **+ friendship**. A score such as "0 vs 0" is kept for the session.

If the other toy does not answer within a few seconds, the calling toy gives up and goes back home. Try again once both toys are on the home screen.

19. Clock, sound and password

The clock screen

Press **CLOCK** to see the time, with seconds and AM/PM. Press **ESC** to go back. To set the time, see chapter 4.

Sound on and off

On the clock screen, **◀** turns the sound **off** (the bell disappears) and **▶** turns it back **on**.

The password lock

After the clock has been set, the toy asks for a **NEW P.W.** (new password): two digits. **◀** changes the first digit, **▶** the second, and **ENTER** confirms.

- **00** means no password. Leave it like that if you don't need one.
- With a password, the toy locks after **4 minutes** without a button press. The next press asks for the password, and only the right one gives access. Handy if you don't want anyone else playing with your twins.

Write your password down. The program does not show it anywhere.

A secret from the programmers. If you sign the name **SCD**, two spaces, then **TEST**, the toy starts in its factory **test mode**. On the home screen (with no menu icon lit) **ENTER** then makes the twins a day older, **ESC** moves the clock on by 10 minutes and a long **ESC** changes their stage. It is fun for seeing the whole life story quickly, but it spoils the real game.

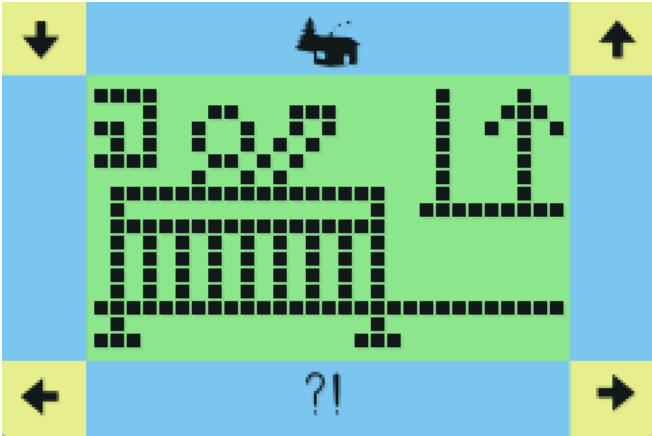
20. Money

A new pair starts with **\$2400**. Money pays for TV and music, the doctor and parties, and a mess in the house costs money too. Work outings bring money in. Keep an eye on the banknote page of the status meter: below \$1000, "?! " warns you, and **at \$0 the twins collapse**.

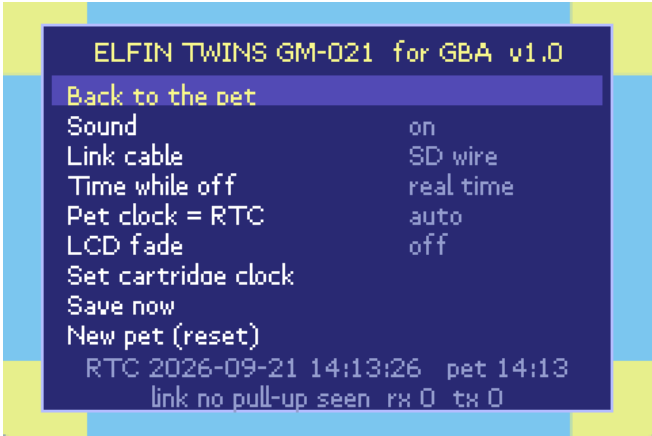
Costs	Earns
TV, music: \$100 · doctor: \$100–200 · parties: \$400 · lessons: \$200 · a mess: \$200 · every hour of illness: \$200 · strong drinks: \$100	Work outings: +\$100 each (hard work +\$300). Emergency rescue resets money to \$2000.

21. Game Boy Advance edition

The GBA edition runs the **original program** of the toy on an emulated copy of its microcontroller, so the twins behave exactly as on the real toy. It adds saving, living on while the GBA is off, a settings menu and the link cable.



On the GBA screen.



START opens the settings.

GBA	Toy button
A	ENTER
B	ESC
+ D-pad	▲ ▼ ◀ ▶
SELECT L R	CLOCK
A + B	ESC+ENTER (rescue / new twins)
START	Settings menu (the twins keep living while it is open)

While the GBA is off

With a flashcart that has a real-time clock, the twins **keep living while the GBA is switched off**. At power-on the game works out how long it has been and lets the twins live through that time in a few seconds, exactly as the toy would have done. So feed them before you switch off.



At power-on after a night away.

Game Boy Advance edition: settings

Setting	What it does
Sound	Buzzer on or off.
Link cable	Which wire of the GBA link cable is used (SD , SC) or off.
Time while off	real time (the twins live on), max 1 day , or paused (nothing happens while off).
Pet clock = RTC	auto sets the toy's clock from the cartridge clock at every start, so you never set it by hand.
LCD fade	Off by default. When on, the dots fade in and out softly, like a slow LCD. The toy's program only draws plain on/off dots, so this is purely for looks.
Set cartridge clock	Set the flashcart's real-time clock.
Save now / New pet	Save immediately / start over.

The game saves by itself a few seconds after you press a button, and once a minute.

Game Boy Advance edition: link cable

- 1 Put **elfin-twins.gba** on both flashcarts and connect the GBAs with a GBA link cable.
- 2 Leave **Link cable** on **SD wire** on both, and play as in chapter 18.
- 3 If the other GBA never answers, run **elfin-linktest.gba** on both first (next section). If it reports "SD idles LOW", switch both games to **SC wire**.

The link and clock test (elfin-linktest.gba)

A small test program for your hardware. Run it on both GBAs with the cable connected:

- **SD / SC idle HIGH**: the wire can be used.
- Press **L** on one GBA: the other should report a **burst of 12 edges** on SD. **R** does the same on SC.
- **A** / **B** hold SD / SC low: the level shown on the other GBA should change.
- **RTC** shows the date and time and should keep ticking. "NOT FOUND" means the flashcart's clock is switched off for this game in the loader menu.

Troubleshooting

Problem	Try this
No time passes while off	Turn on the RTC for this game in the flashcart menu. Check "Time while off".
"no pull-up seen" in the menu	Nothing seems to be connected, or the wire doesn't idle high. Check the cable, then try SC.
The pet is not saved	Use SRAM saving for this game in the flashcart menu. Some flashcarts only write the save to the SD card at the next start.
The pet clock is wrong	Set the cartridge clock (settings), and use "Pet clock = RTC: auto".

22. Who made Elfin Twins?

No box, leaflet or label naming the maker of the GM-021 has turned up. But the program inside the toy has a signature. The last bytes before the end of its memory hold one line of text, which the toy never shows:

Sparkle Crystal Development Co.,Ltd. WangLaXian 1996.11.16

stored bytes, and each byte minus \$20:

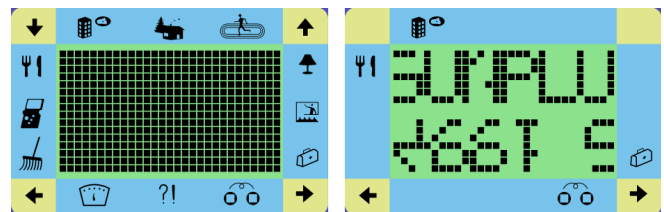
\$FF27	73	90	81	92	8B	8C	85	40	63	92	99	93	94	81	8C	40	64	85	96	85
	S	p	a	r	k	l	e			C	r	y	s	t	a	l		D	e	v
\$FF3B	8C	8F	90	8D	85	8E	94	40	63	8F	4E	4C	6C	94	84	4E	40	77	81	8E
	l	o	p	m	e	n	t			C	o	.	,	L	t	d	.		W	a
\$FF4F	87	6C	81	78	89	81	8E	40	51	59	59	56	4E	51	51	4E	51	56		
	g	L	a	X	i	a	n			1	9	9	6	.	1	1	.	1	6	

The credit as stored in the toy. Each letter is saved as its normal computer code plus 32, so the text does not show up when you search the memory for words. Subtract 32 and it reads plainly.

- **Sparkle Crystal Development Co., Ltd.** is most likely the company that wrote the program. The toy's secret test name **SCD TEST** (chapter 19) matches its initials: **Sparkle Crystal Development**.
- **WangLaXian** looks like a Chinese personal name written in pinyin (Wang La-xian), almost certainly the programmer who signed the work.
- **16 November 1996** is probably when the program was finished: about a week before the first Tamagotchi went on sale in Japan, and months before the wave of virtual pets of 1997.

A chip from Sunplus

The toy's brain is a Sunplus SPLB20 chip. Its first kilobyte holds Sunplus's own factory test program, the same in every SPLB20 toy we have looked at. At the factory, a tester could use it to check the screen, the buzzer, the memory and the program itself before the chip went into a toy. One of its screen tests even writes **SUNPLU...** across the display.



Factory screen tests: all segments on (left) and Sunplus's test picture (right), recorded by running the test program on the emulator.

Still unknown. Nothing about Sparkle Crystal Development could be found online: not its location, its other products, or what became of it. Of the thirty-odd toys dumped for the BrickEmuPy emulator, only Elfin Twins carries this signature. *Apollo Prince & Princess* stores its letters in exactly the same way, and 16 of its 26 letters are identical, but that is not enough to say it came from the same studio. If you own an Elfin Twins in its box, the packaging may name the distributor. Company registries (for example Hong Kong's) could show when the firm existed.

23. Quick reference

Menu (◀ ▶ then ENTER)

-  **Food & drink:** meals left/right, drink in the middle
-  **Games:** find the heart, lucky reels
-  **Clean:** when dirty or messy
-  **Status meter:** 8 pages
-  **Link:** play with a friend
-  **Doctor:** when ill
-  **Outings:** TV, music, reading, work, sport...
-  **Light:** bedtime after 22:00

Buttons

- ▲ choose a destination · ▼ cancel
- CLOCK** clock · hold: stop / set the clock
- On the clock: ◀ sound off · ▶ sound on
- Setting: ◀ hour +1 · ▶ minute +1 · **ENTER** start
- ESC+ENTER:** rescue / new twins

The twins' day

- 08:00 up, hungry and thirsty
- Out: before 18:00, back by 20:00
- 22:00 sleepy: light off
- Light off on time: +1 kg · 23:00 they fall asleep anyway: -1 kg
- 07:00 the alarm rings

Life

- Babies days 1–3 · children 4–8 · teenagers 9–15 · adults 16+
- A birthday every 24 h
- Stay: 23–26 days
- Collapse: ill too long, or \$0
- Rescue: ESC+ENTER within 24 h

Elfin Twins GM-021 manual, reconstructed in 2026 from the toy's program with the BrickEmuPy emulator (reverse engineering of the device by azya52; device provided by t.me/oksiniah) and the Game Boy Advance port in this repository.

Elfin Twins is the property of its respective owners; this is an unofficial fan document.